Lab Report: 3

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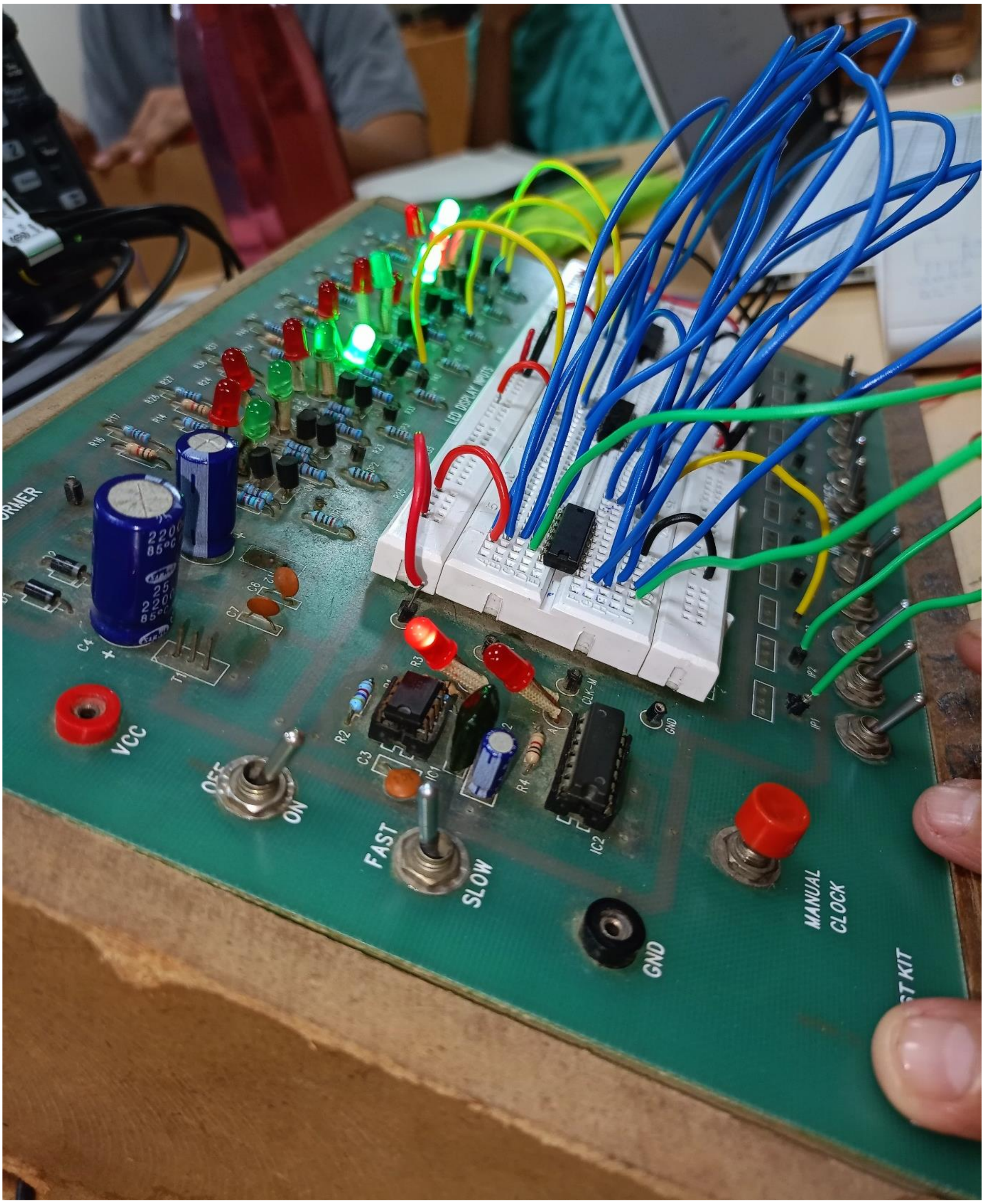


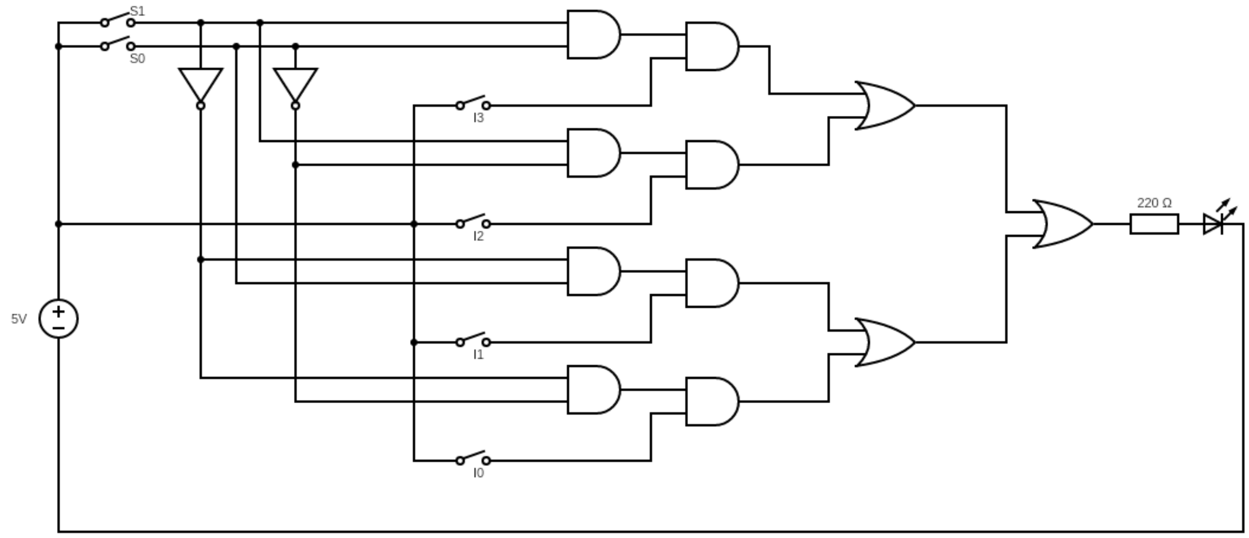
Part-A

Objective: Designing 4:1 Multiplexer using basic logic gates

Electronic components used:

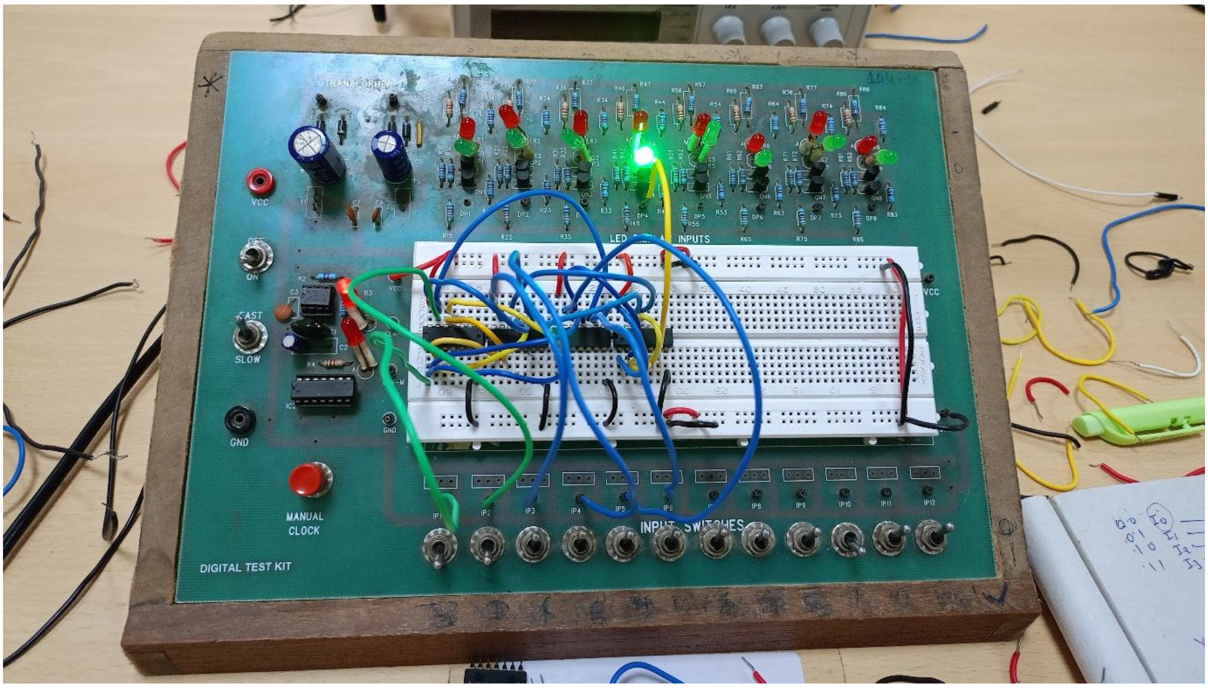
- Digital test kit (w/ breadboard)
- 1x 74HC04 IC
- 2x 74HC11 ICs
- 1x 74HC32 IC
- Wires

The reference circuit:



Procedure:

- Set up the circuit on the breadboard as shown in the reference diagram.
- Observe outputs on the LED against all possible inputs through the toggles.
- Tabulate the observations into a truth table.



Conclusion: The truth table of a 4-to-1 multiplexer came out as:

S1	S0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Link for the Tinkercad simulation:

https://www.tinkercad.com/things/fIdQ2UMQfIx-l3parta/editel?sharecode=LDQL6a472cgXzCia6OoH2pR0e6nXj5h6j65p_Yz2s9Y

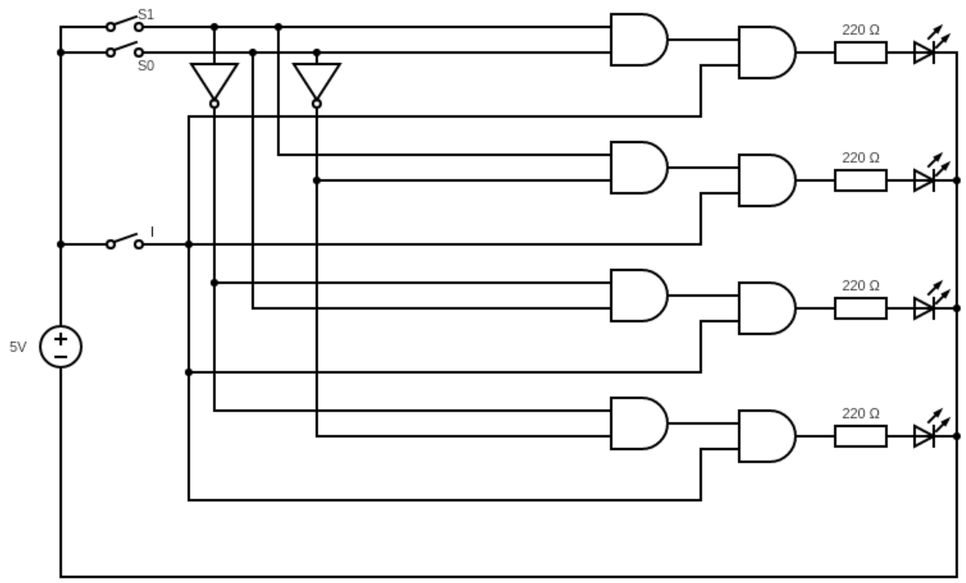
Part-B

Objective: Designing 1:4 Demultiplexer using basic logic gates

Electronic components used:

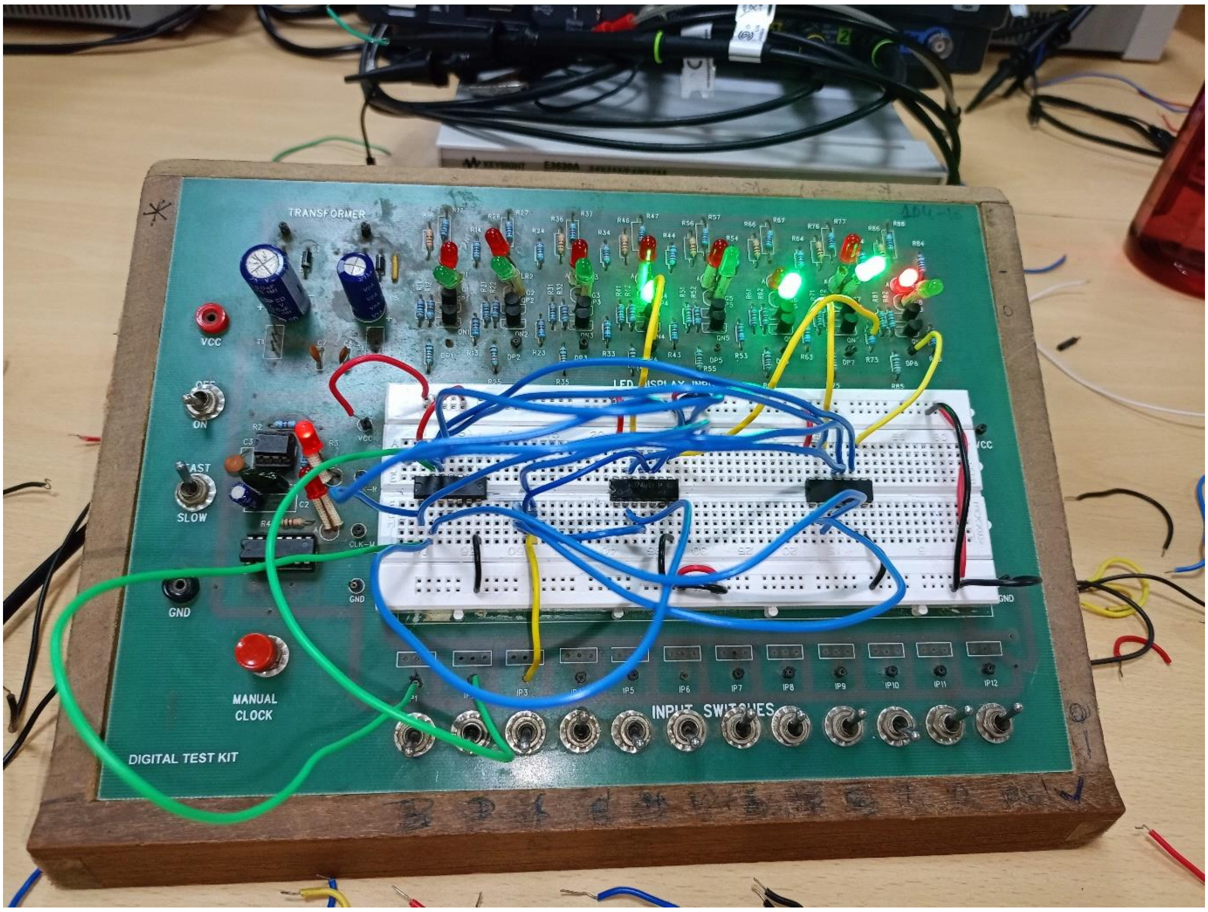
- Digital Test Kit (w/ breadboard)
- 1x 74HC04 IC
- 2x 74HC11 ICs
- Wires

The reference circuit:



Procedure:

- Set up the circuit on the breadboard as shown in the reference diagram.
- Observe outputs on the LEDs against all possible inputs through the toggles.
- Tabulate the observations into a truth table.



Conclusion: The truth table of a 4-to-1 multiplexer came out as:

S1	S0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0

Link for the Tinkercad simulation:

https://www.tinkercad.com/things/3RcMGpTZToM-l3partb/editel?sharecode=M_ennQyRwCt35bX9hbljiU04xZV4-pmhyoohJZpRCyg

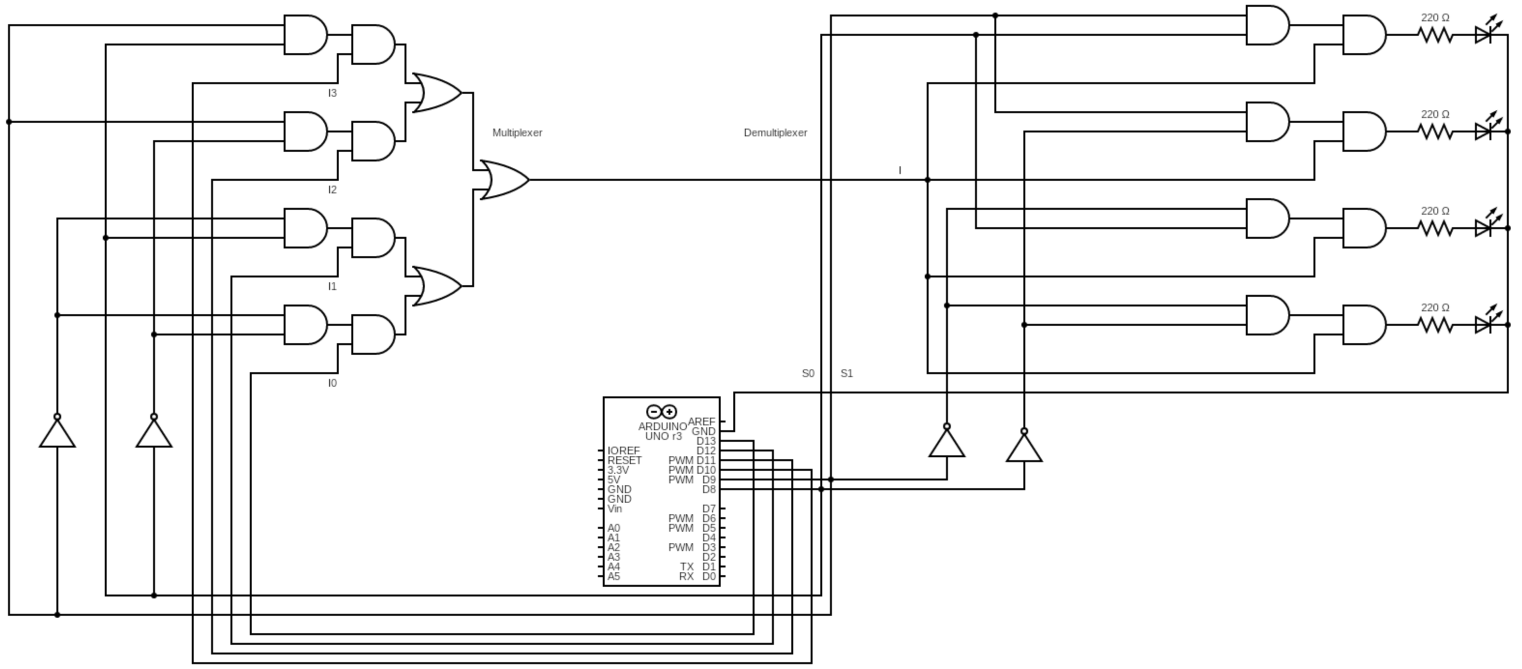
Part-C

Objective: Assembling the Mux-Demux circuit and testing it with Arduino

Electronic components used:

- Digital test kit (w/ breadboard)
- Part-A circuit
- Part-B circuit
- Arduino Uno
- Wires

The reference circuit:

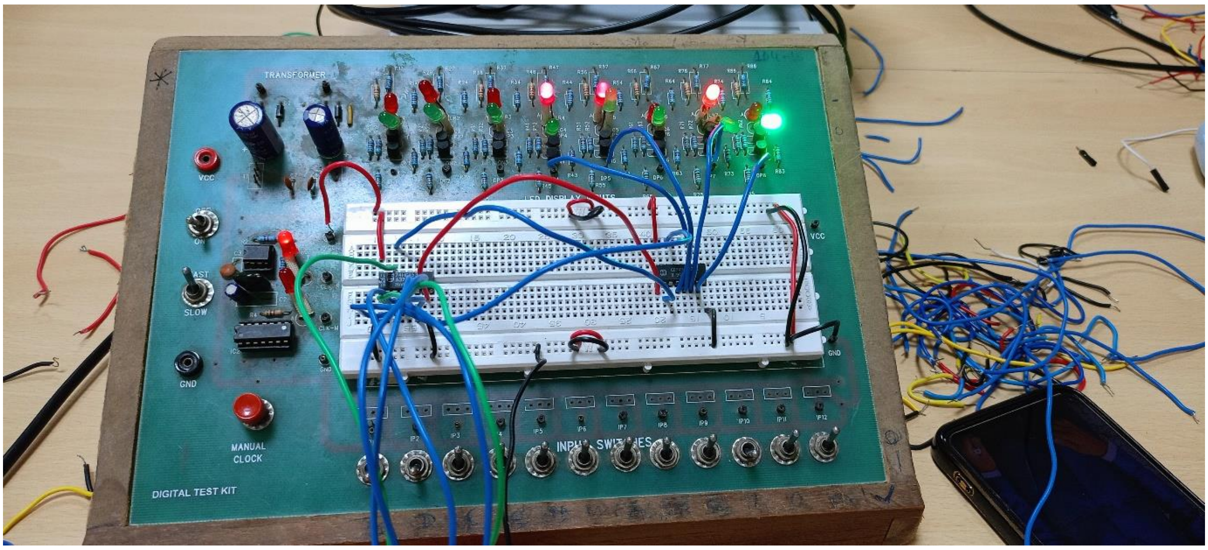


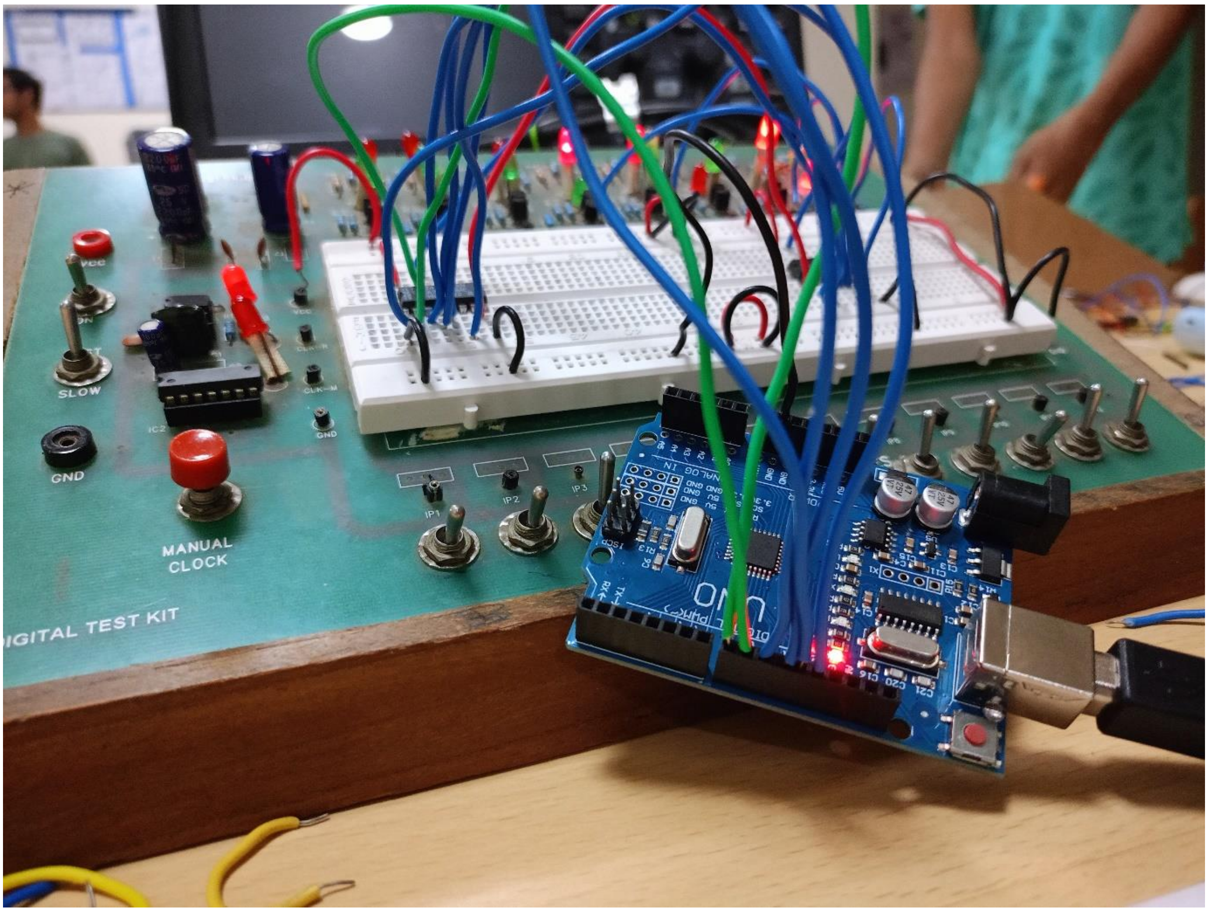
Arduino Code:

```
//Auth: @Saarthak_2023102055
//Mux-Demux Controller
//Accepts serial input in the format:
//  S0|S1|I0|I1|I2|I3 0|1
//to set the values for Select/Input switches

int S[] = {9, 8};
int I[] = {13, 12, 11, 10};
void setup()
{
  Serial.begin(9600);
  for(int i = 8; i <= 13; i++)
    pinMode(i, OUTPUT);
}

void loop()
{
  while(!Serial.available());

  String cmd = Serial.readString();
  Serial.println(cmd);

  switch(cmd[0]) {
    case 'S':
        digitalWrite(S[cmd[1]-'0'], cmd[3]-'0');
        break;
    case 'I':
        digitalWrite(I[cmd[1]-'0'], cmd[3]-'0');
        break;
  }
}
```

Procedure:

- Connect the final output of the multiplexer to the input of the demultiplexer instead of the LED.
- Replace the data inputs to the multiplexer with pins 13, 12, 11 and 10.
- Replace the select inputs to both the circuits with pins 9 and 8.
- Remove the power supply and ground the common cathode of the demultiplexer output LEDs using the GND pin.
- Compile and upload the program shown to Arduino.
- Provide inputs through the pins using the Serial Monitor.





Conclusion: The assembled Mux-Demux circuit came out to be functional and consistent.

Link for the Tinkercad simulation:

https://www.tinkercad.com/things/3wlBNjSlsx9-l3partc/editel?sharecode=wI2vTp3qOTJA79cqttY1ZsQxW2ST3D5nbn3KhC5nW7Y